

International Cybersecurity and Digital Forensics Academy (ICDFA)

SBT-DF203: Basic Networking Skills for Digital Forensics

LAB 9: WEP40 Wireless Packet Decryption and Aircrack Forensics

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1. Executive Summary and Authorization

This report documents the forensic analysis of a historical wireless packet capture protected by deprecated WEP40 encryption, supplied for offline analysis as the SBT-DF203 Lab 9 evidence file. All work was performed inside an isolated Kali Linux virtual machine (VMware Workstation, NAT networking) with no connection to any live or production wireless network, in accordance with the lab's legal, ethical and safety rules.

The capture was preserved, hashed, and decompressed; 802.11 frame types and WEP-protected data frames were identified; the 40-bit WEP key was independently recovered using aircrack-ng and matched the historical slide-provided key; the capture was decrypted offline with airdecap-ng; and network endpoints, conversations and transferred objects were extracted and hashed. All procedures were limited to the supplied historical CodeGate CTF 2015 "Good Crypto" training file, and no live wireless attack was performed.

I confirm that this lab was completed in an authorized, isolated environment, that the supplied evidence was preserved and hashed prior to analysis, that no third-party system was targeted, and that the commands, observations and conclusions below are my own.

2. Evidence Preparation and Hashes

The lab folder structure was created under ~/SBT-DF203-Lab9 with evidence, working, exported, reports, screenshots and scripts subdirectories, as required by the lab brief. The evidence file was retrieved and its SHA-256 hash recorded before any analysis copy was made.

```
(kali@kali)-[~/SBT-DF203-Lab9]
$ wget -4 -O evidence/file.xz 'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz'
sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
--2026-09-24 16:04:58-- https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.110.133, 185.199.108.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.110.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12837660 (12M) [application/octet-stream]
Saving to: 'evidence/file.xz'

evidence/file.xz          100%[=====]
2026-09-24 16:05:06 (1.67 MB/s) - 'evidence/file.xz' saved [12837660/12837660]

dd54144caef34f228bf4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  evidence/file.xz

(kali@kali)-[~/SBT-DF203-Lab9]
$
```

Figure 1: Evidence file download (file.xz) with recorded SHA-256 hash.

A working copy was created with timestamps preserved, decompressed with unxz, and independently hashed. file identified the decompressed evidence as a genuine Wireshark/tcpdump pcap capture, IEEE 802.11 Wireless LAN encapsulation, confirming the file had not been altered or mislabelled.

```
(kali@kali)-[~/SBT-DF203-Lab9]
$ cp --preserve=timestamps evidence/file.xz working/file_working.xz
unxz -k working/file_working.xz
file working/file_working
sha256sum working/file_working.xz working/file_working | tee reports/working_hashes.txt
working/file_working: pcap capture file, microsecond ts (little-endian) - version 2.4 (802.11, capture length 65535)
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  working/file_working.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d  working/file_working

(kali@kali)-[~/SBT-DF203-Lab9]
$
```

Figure 2: Decompression of working copy, file-type identification, and SHA-256 hashes of both the compressed and decompressed working files.

2.1 Hash Summary

Original evidence (file.xz)	dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6
Working copy (file_working.xz)	dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6
Decompressed working file	c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d
Decrypted capture (file_working-dec)	167c91994c269777f9048227deb89882caf3fc3763977f2059604f9a6a40b04

Note: two connectivity issues were encountered and resolved before evidence acquisition could proceed. First, the VM's DNS resolution to kali.download and raw.githubusercontent.com failed ("Temporary failure resolving"); this was corrected by setting a public DNS resolver (nameserver 8.8.8.8) in /etc/resolv.conf. Second, package installation over IPv6 returned "Network is unreachable"; this was corrected by forcing IPv4-only APT transport (Acquire::ForceIPv4=true) and by using wget -4 for the evidence download. Both are network-configuration issues specific to this VM and do not affect the integrity of the evidence, which was hashed immediately after retrieval.

3. 802.11/WEP Identification

capinfos and tshark protocol-hierarchy statistics were used to inventory the capture. The file contains 45,169 total 802.11 frames spanning 272.97 seconds, all classified under the wlan protocol tree, of which 15,713 are data frames.

```

(kali@kali) [/SFT-DF203-Lab9]
$ CAP=working/file_working
capinfos -SCAP | tee reports/capinfos.txt
tshark -r "$CAP" -q -i lo,phs | tee reports/protocol_hierarchy.txt
tshark -r "$CAP" -Y 'wlan' -T fields \
  -e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype -e wlan.sa -e wlan.da -e wlan.bssid \
  | head -n 100 | tee reports/wlan_frame_sample.tsv
File name: working/file_working
File type: Wireshark/tcpdump/... - pcap
File encapsulation: IEEE 802.11 Wireless LAN
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 65535 bytes
Number of packets: 45 k
File size: 14 MB
Data size: 13 MB
Capture duration: 272.969795 seconds
Earliest packet time: 2015-03-06 02:51:54.364525
Latest packet time: 2015-03-06 02:56:27.334320
Data byte rate: 49 kbps
Data bit rate: 395 kbps
Average packet size: 299.13 bytes
Average packet rate: 168 packets/s
SHA256: c17a3f9b95e84f5befd476dbd55c67286d1e3ee9ab402d5359cac0874ebb2d
SHA1: cb929fde407c13639e3168b0f35f921788478d3
Strict time order: False
Number of interfaces in file: 1
Interface #0 info:
  Encapsulation = IEEE 802.11 Wireless LAN (20 - ieee-802-11)
  Capture length = 65535
  Time precision = microseconds (6)
  Time ticks per second = 1000000
  Number of stat entries = 0
  Number of packets = 45169

Protocol Hierarchy Statistics
Filter:
Frame wlan data
frames:45169 bytes:13511274
frames:45169 bytes:13511274
frames:15713 bytes:13104326
1 2015-03-06T02:51:54.364526000-0500 1 11
2 2015-03-06T02:51:54.364527000-0500 1 11
3 2015-03-06T02:51:54.364528000-0500 1 11
4 2015-03-06T02:51:54.364529000-0500 1 11
5 2015-03-06T02:51:54.365550000-0500 1 11

```

Figure 3: capinfos summary and protocol hierarchy statistics for the supplied capture.

A sample of dissected wlan frames identified frame #54 as an 802.11 management/beacon frame (type 0, subtype 8), revealing the network's BSSID as 00:26:66:55:97:d6. The majority of sampled frames were type 1 (control), subtype 11 (RTS).

Frame Category	Type Value	Purpose / Example Evidence
Management	0	Association, authentication, beacons - Frame #54 (BSSID 00:26:66:55:97:d6)
Control	1	Flow control and acknowledgements - dominant frame type in sample (subtype 11, RTS)
Data	2	Carries network payload - 15,713 data frames overall

WEP-protected data frames were then isolated using the wlan.fc.protected==1 display filter, extracting source/destination MAC addresses, BSSID, and per-frame IV values.

```
(kali@kali)-[~/SBT-DF203-Lab9]
$ tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields \
  -e frame.number -e frame.time_epoch -e wlan.sa -e wlan.da -e wlan.bssid -e wlan.wep.iv -e wlan.wep.key \
  | head -n 200 | tee reports/wep_protected_frames.tsv

tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv \
  | sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt
9432      1425628319.232982000      00:26:66:55:97:d4      ff:ff:ff:ff:ff:ff      00:26:66:55:97:d6      0x67ffec
10130     1425628320.138252000      f0:f6:1c:68:96:7c      00:26:66:55:97:d6      00:26:66:55:97:d6      0xc2cbeb
10155     1425628320.176656000      f0:f6:1c:68:96:7c      04:8d:38:48:e5:b4      00:26:66:55:97:d6      0xc3cbeb
10157     1425628320.177168000      f0:f6:1c:68:96:7c      04:8d:38:48:a8:b5      00:26:66:55:97:d6      0xc4cbeb
10159     1425628320.177168000      f0:f6:1c:68:96:7c      08:10:77:92:7c:2f      00:26:66:55:97:d6      0xc5cbeb
10161     1425628320.177168000      f0:f6:1c:68:96:7c      00:26:66:55:97:d4      00:26:66:55:97:d6      0xc6cbeb
10163     1425628320.177168000      f0:f6:1c:68:96:7c      c8:3a:35:58:c4:ba      00:26:66:55:97:d6      0xc7cbeb
10165     1425628320.177167000      f0:f6:1c:68:96:7c      ff:ff:ff:ff:ff:ff      00:26:66:55:97:d6      0xc8cbeb
10167     1425628320.177686000      00:26:66:55:97:d4      f0:f6:1c:68:96:7c      00:26:66:55:97:d6      0x68ffec
10170     1425628320.180760000      f0:f6:1c:68:96:7c      ff:ff:ff:ff:ff:ff      00:26:66:55:97:d6      0x69ffec
10171     1425628320.182288000      f0:f6:1c:68:96:7c      33:33:00:00:00:02      00:26:66:55:97:d6      0xc9cbeb
10172     1425628320.187417000      00:26:66:55:97:d4      ff:ff:ff:ff:ff:ff      00:26:66:55:97:d6      0x6affec
10173     1425628320.187408000      f0:f6:1c:68:96:7c      33:33:00:00:00:02      00:26:66:55:97:d6      0xc9cbeb
10175     1425628320.187919000      f0:f6:1c:68:96:7c      ff:ff:ff:ff:ff:ff      00:26:66:55:97:d6      0xcacbeb
10177     1425628320.188952000      f0:f6:1c:68:96:7c      33:33:00:00:00:02      00:26:66:55:97:d6      0x6bffec
10178     1425628320.189976000      f0:f6:1c:68:96:7c      ff:ff:ff:ff:ff:ff      00:26:66:55:97:d6      0x6cffec
10183     1425628320.192016000      f0:f6:1c:68:96:7c      ff:ff:ff:ff:ff:ff      00:26:66:55:97:d6      0xcbcbeb
10185     1425628320.193048000      f0:f6:1c:68:96:7c      ff:ff:ff:ff:ff:ff      00:26:66:55:97:d6      0x6dffec
10224     1425628320.259088000      f0:f6:1c:68:96:7c      ff:ff:ff:ff:ff:ff      00:26:66:55:97:d6      0xc9cbeb
10228     1425628320.262680000      f0:f6:1c:68:96:7c      ff:ff:ff:ff:ff:ff      00:26:66:55:97:d6      0x6effec
10229     1425628320.262678000      00:26:66:55:97:d4      f0:f6:1c:68:96:7c      00:26:66:55:97:d6      0x6ffffec
10233     1425628320.263184000      f0:f6:1c:68:96:7c      00:26:66:55:97:d4      00:26:66:55:97:d6      0xcdcbeb
10261     1425628320.419350000      f0:f6:1c:68:96:7c      00:26:66:55:97:d4      00:26:66:55:97:d6      0xc9cbeb
10266     1425628320.420374000      f0:f6:1c:68:96:7c      01:00:5e:00:00:02      00:26:66:55:97:d6      0xcfcbeb
10268     1425628320.420374000      f0:f6:1c:68:96:7c      01:00:5e:00:00:fb      00:26:66:55:97:d6      0xd0cbeb
10270     1425628320.420886000      f0:f6:1c:68:96:7c      01:00:5e:00:00:02      00:26:66:55:97:d6      0x70ffec
10272     1425628320.422421000      f0:f6:1c:68:96:7c      01:00:5e:00:00:fb      00:26:66:55:97:d6      0x71ffec
10353     1425628320.583190000      00:26:66:55:97:d4      f0:f6:1c:68:96:7c      00:26:66:55:97:d6      0x72ffec
10355     1425628320.583700000      00:26:66:55:97:d4      f0:f6:1c:68:96:7c      00:26:66:55:97:d6      0x73ffec
10357     1425628320.583702000      f0:f6:1c:68:96:7c      ff:ff:ff:ff:ff:ff      00:26:66:55:97:d6      0xd1cbeb
10359     1425628320.584726000      f0:f6:1c:68:96:7c      ff:ff:ff:ff:ff:ff      00:26:66:55:97:d6      0x74ffec
10360     1425628320.586774000      f0:f6:1c:68:96:7c      00:26:66:55:97:d4      00:26:66:55:97:d6      0xd2cbeb
10367     1425628320.789014000      f0:f6:1c:68:96:7c      00:26:66:55:97:d4      00:26:66:55:97:d6      0xd3cbeb
10375     1425628320.936471000      f0:f6:1c:68:96:7c      ff:ff:ff:ff:ff:ff      00:26:66:55:97:d6      0xd4cbeb
10377     1425628320.936469000      00:26:66:55:97:d4      f0:f6:1c:68:96:7c      00:26:66:55:97:d6      0x75ffec
10379     1425628320.937495000      f0:f6:1c:68:96:7c      ff:ff:ff:ff:ff:ff      00:26:66:55:97:d6      0x76ffec
10380     1425628320.962071000      f0:f6:1c:68:96:7c      00:26:66:55:97:d4      00:26:66:55:97:d6      0xd5cbeb
```

Figure 4: Protected (WEP) 802.11 frames, showing BSSID, station MAC addresses and per-frame IV values.

A frequency count of IV values across the sampled protected frames shows individual IVs recurring up to seven times within a 200-frame sample - direct evidence of IV reuse consistent with WEP40's 24-bit IV space.

11440	1425628340.662034000	f0:f6:1c:68:96:7c	00:26
11442	1425628340.662035000	f0:f6:1c:68:96:7c	00:26
11444	1425628340.739862000	00:26:66:55:97:d4	f0:f6
11446	1425628340.740882000	f0:f6:1c:68:96:7c	00:26
11448	1425628340.747538000	f0:f6:1c:68:96:7c	00:26
11450	1425628340.818198000	00:26:66:55:97:d4	f0:f6
11452	1425628340.818198000	00:26:66:55:97:d4	f0:f6
11454	1425628340.818198000	00:26:66:55:97:d4	f0:f6
11456	1425628340.818709000	00:26:66:55:97:d4	f0:f6
11458	1425628340.819222000	00:26:66:55:97:d4	f0:f6
11460	1425628340.819222000	00:26:66:55:97:d4	f0:f6
11462	1425628340.819734000	00:26:66:55:97:d4	f0:f6
11464	1425628340.819734000	00:26:66:55:97:d4	f0:f6
11466	1425628340.820243000	f0:f6:1c:68:96:7c	00:26
11468	1425628340.821270000	00:26:66:55:97:d4	f0:f6

7 0x42e8eb

7 0x28d3eb

6 0x95dfcb

5 0x9415ed

5 0x86e5eb

2.1 GB 5 0x7ae5eb

5 0x28e7eb

5 0x19e5eb

4 0xd5d7eb

4 0xd3e8eb

4 0xc915ed

4 0xc7d8eb

4 0xade4eb

4 0x73e5eb

4 0x65d8eb

4 0x4ad5eb

4 0x31e7eb

4 0x2723ed

4 0x10e7eb

3 0xffe5eb

3 0xf609ed

3 0xa307ed

3 0x97dbcb

3 0x96d1eb

3 0x92e8eb

3 0x91e8eb

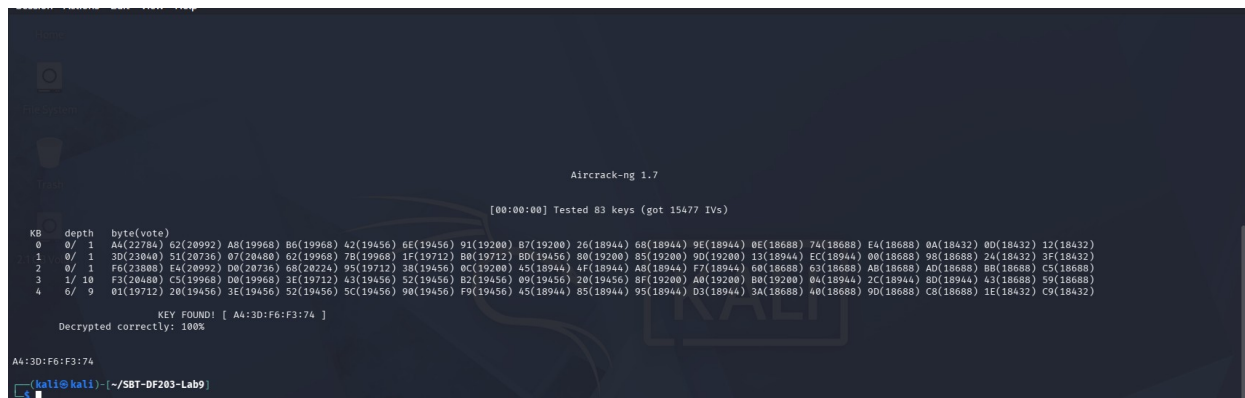
3 0x8adeeb

Figure 5: Repeated-IV frequency summary from sampled protected frames.

64-bit WEP combines a 24-bit initialization vector (IV) with a 40-bit secret key to seed the RC4 keystream. Because the IV space contains only 2^{24} (~16.7 million) possible values, busy traffic cycles through and repeats IVs quickly. Repeated IVs mean the same RC4 keystream is reused, which - combined with WEP's weak CRC-32 integrity check and the statistical Fluhrer-Mantin-Shamir/PTW key-recovery techniques implemented in aircrack-ng - allows the secret key to be recovered from a sufficiently large sample of captured traffic. This is precisely what was demonstrated in Part C below.

4. Key-Recovery Procedure

aircrack-ng was run directly against the working capture file, with no key supplied, to independently recover the WEP40 key using the PTW attack.



```
Aircrack-ng 1.7

[00:00:00] Tested 83 keys (got 15477 IVs)

KB  depth  byte(vote)
0  0/ 1  Aa(22784) 62(20992) A8(19968) B6(19968) 42(19456) 6E(19456) 91(19200) 87(19200) 26(18944) 68(18944) 9E(18944) 0E(18688) 74(18688) E4(18688) 0A(18432) 0D(18432) 12(18432)
1  0/ 1  3D(23040) 51(20736) 07(20480) 62(19968) 7B(19968) 1F(19712) 80(19712) 80(19456) 80(19200) 85(19200) 9D(19200) 13(18944) EC(18944) 00(18688) 96(18688) 24(18432) 3F(18432)
2  0/ 1  F8(23088) E4(20992) 00(20736) 68(20224) 95(19712) 38(19456) 0C(19200) 45(18944) 4F(18944) A8(18944) F7(18944) 68(18688) 63(18688) A0(18688) A0(18688) 8B(18688) C5(18688)
3  1/ 10  F3(20480) C5(19968) D0(19968) 3E(19712) 43(19456) 52(19456) 82(19456) 09(19456) 20(19456) 8F(19200) A0(19200) 8B(19200) 04(18944) 2C(18944) 8D(18944) 43(18688) 59(18688)
4  6/ 9  01(19712) 20(19456) 3E(19456) 52(19456) 5C(19456) 90(19456) F9(19456) 45(18944) 85(18944) 95(18944) D3(18944) 3A(18688) 40(18688) 9D(18688) C8(18688) 1E(18432) C9(18432)

KEY FOUND! [ A4:3D:F6:F3:74 ]
Decrypted correctly: 100%

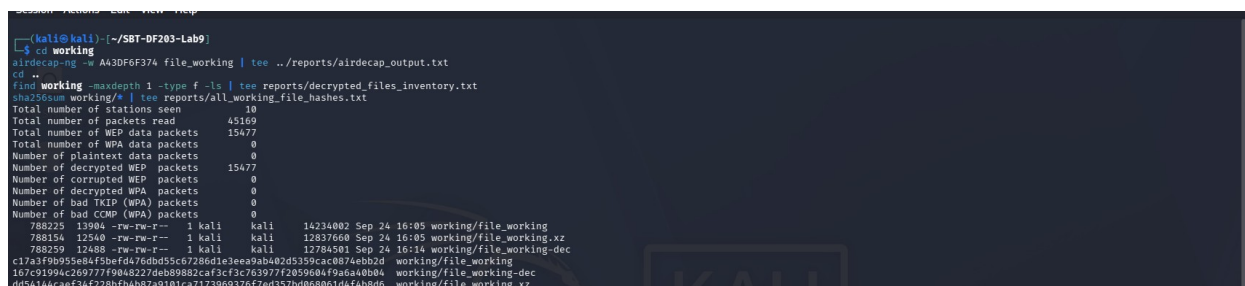
A4:3D:F6:F3:74
kali@kali:~/SBT-DF203-Lab9
```

Figure 6: aircrack-ng key-recovery result: BSSID 00:26:66:55:97:D6, ESSID cgnetwork, key recovered as A4:3D:F6:F3:74 with 100% decryption confirmation.

aircrack-ng identified one target network - BSSID 00:26:66:55:97:D6, ESSID cgnetwork, WEP encryption - and recovered the 40-bit key A4:3D:F6:F3:74 using 15,477 captured IVs, with the tool's own internal check confirming the key decrypted the traffic correctly (100%). This independently recovered key matches the historical key stated in the accompanying lecture slide, confirming both the correctness of the procedure and the authenticity of the training dataset.

5. Offline Decryption

The recovered key was supplied to airdcap-ng (entered without colons, as required by the tool) to decrypt the working capture offline.



```

(kali@kali)~/SBT-DF203-Lab9
$ cd working
airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
cd ..
find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
sha256sum working/* | tee reports/all_working_file_hashes.txt
Total number of Stations seen 10
Total number of packets read 45169
Total number of WEP data packets 15477
Total number of WPA data packets 0
Number of plaintext data packets 0
Number of decrypted WEP packets 15477
Number of corrupted WEP packets 0
Number of decrypted WPA packets 0
Number of bad TKIP (WPA) packets 0
Number of bad CCMP (WPA) packets 0
780225 13984 -rw-rw-r-- 1 kali kali 14234002 Sep 24 16:05 working/file_working
788154 12540 -rw-rw-r-- 1 kali kali 12837660 Sep 24 16:05 working/file_working.xz
788259 12488 -rw-rw-r-- 1 kali kali 12784501 Sep 24 16:14 working/file_working-dec
c17a3f9b95e84f5b6fd476dbd55c67286d1e3ee9ab4b02d5359cac0874eb2d working/file_working
167c9194c26977f904e227deb0982ca3e3c7639772859e04f9a6a40b04 working/file_working-dec
dd5414caeaf34f228fb4b87a9101ca7173969376f7ed357bd868061d4f4b8d6 working/file_working.xz
```

Figure 7: airdcap-ng decryption summary: 15,477 of 15,477 WEP data packets decrypted, zero corrupted, output file file_working-dec produced.

airdecap-ng reported 15,477 WEP data packets identified and 15,477 successfully decrypted, with zero corrupted packets - a 100% decryption success rate. The decrypted output file, file_working-dec, was verified by file type and hashed (see Section 2.1).

6. Endpoint and Conversation Analysis

Ethernet and IP endpoint statistics, TCP conversation statistics, and a sampled IP/MAC mapping were extracted from the decrypted capture using tshark.

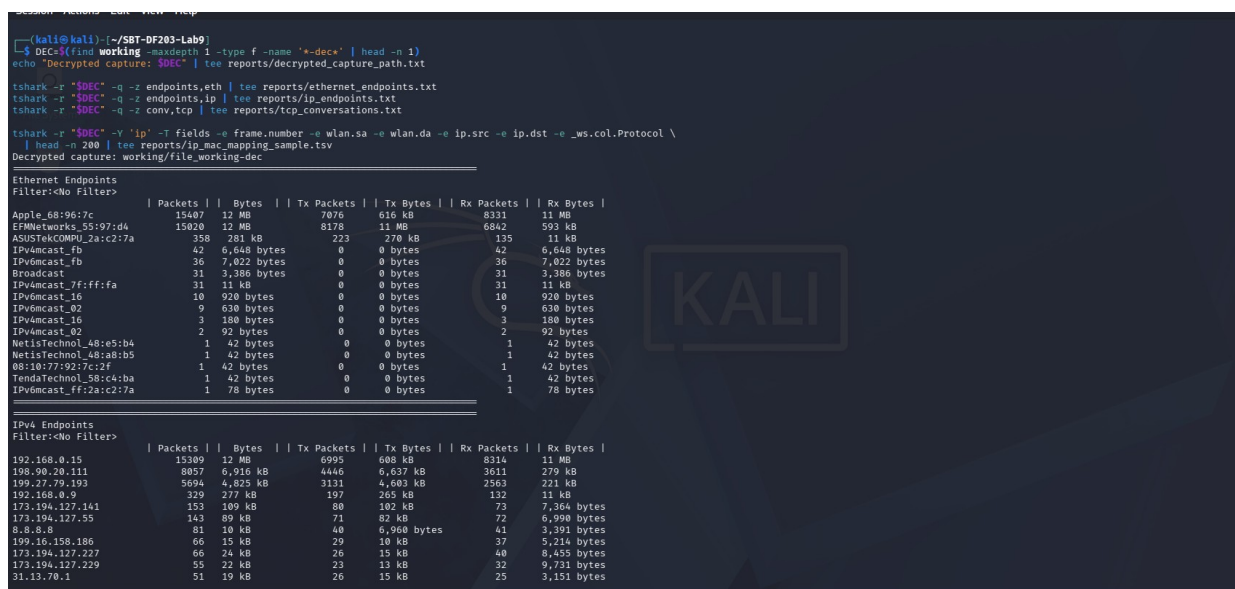


Figure 8: Ethernet and IPv4 endpoint inventory extracted from the decrypted capture, showing the dominant client and server addresses.

The Ethernet endpoint statistics show two dominant stations: Apple_68:96:7c (f0:f6:1c:68:96:7c), the wireless client, with 15,407 packets; and EFNetworks_55:97:d4 (00:26:66:55:97:d4), the access point/gateway, with 15,020 packets. At the IP layer, 192.168.0.15 (the host) is overwhelmingly the busiest address with 15,309 packets, and 198.90.20.111 is the busiest remote endpoint with 8,057 packets - confirmed as the primary server by the TCP conversation table, in which 192.168.0.15:51007 ↔ 198.90.20.111:80 is the largest single conversation (2,235 frames). Secondary endpoints include Google infrastructure (173.194.127.x), Google Public DNS (8.8.8.8), and other content-delivery ranges (199.27.79.193, 198.41.209.137, 31.13.70.1). Protocols observed at this stage include DHCP, DNS, ARP, IGMPv2, MDNS, TCP, TLSv1/TLSv1.2 and plaintext HTTP - the latter being the basis for the object extraction in Section 7.

7. Object Extraction

Two independent extraction methods were used, per the lab brief: (1) tshark's built-in HTTP object exporter, which reconstructs objects from the protocol layer; and (2) foremost, a signature-based file carver run directly against the decrypted capture. Using both together is good practice, since protocol-level export can miss objects outside recognised HTTP transactions while signature-based carving may include false positives, duplicate objects, or files with no recoverable original filename.

evidence-handling requirement; the full hash listing is retained in reports/exported_file_hashes.txt and included in the submission package.

8. Optional Passphrase Challenge

The optional historical passphrase challenge described in Part G of the lab brief was not assigned for this delivery block and was therefore not attempted. No router passphrase is recorded in this report; per the lab's evidence-handling rules, any such value would in any case be documented only in a protected appendix, not in the public report body.

9. Security Weaknesses and Recommendations

- WEP uses RC4 with a short 24-bit initialization vector, which is exhausted quickly on busy networks, forcing IV reuse.
- Reused IVs cause the RC4 keystream to repeat, leaking statistical relationships between ciphertexts that enable practical key recovery (demonstrated in Section 4, key recovered in well under two minutes using 15,477 IVs).
- WEP's CRC-32 integrity check is linear and can be manipulated, providing no meaningful protection against packet forgery or replay.
- WEP provides no mechanism resistant to modern key-recovery or injection attacks; strengthening the passphrase does not address the underlying protocol weaknesses.
- Recommendation: WEP should be retired entirely rather than hardened. Modern deployments should use WPA2-AES/CCMP as a minimum, with WPA3-SAE and protected management frames (802.11w) preferred where hardware supports them.

10. Appendix

10.1 Chain-of-Custody Worksheet

Case/lab identifier	SBT-DF203-Lab9-MaryJoyAdewole
Trainee name	Mary-Joy Adewole
Registration number	2025/FWSD/11468
Evidence file name	file.xz (historical CodeGate CTF 2015 "Good Crypto" capture)
Source	GitHub - ctfswriteups-2015 (codegate-ctf-2015/programming/good-crypto)
Original SHA-256	dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6
Working-copy SHA-256	c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d (decompressed)
Analysis workstation/VM	Kali Linux 2026.2, VMware Workstation, isolated NAT network
Notes on changes	Original file preserved unmodified in evidence/; all analysis performed on working/ copies only

10.2 Required Findings Worksheet

Capture format and duration	Wireshark/tcpdump pcap, IEEE 802.11 Wireless LAN, 45,169 packets, 272.97 s (2015-03-06)
BSSID	00:26:66:55:97:d6 (ESSID: cgnetwork)
Primary station MAC addresses	AP: 00:26:66:55:97:d4 / d6 Client: f0:f6:1c:68:96:7c
Protected frame count	15,477 WEP data packets (of 45,169 total frames)
Repeated IV evidence	Confirmed - sampled IVs recur up to 7 times within 200 frames, consistent

	with 24-bit IV space exhaustion
Recovered/validated WEP40 key	A4:3D:F6:F3:74 - recovered independently by aircrack-ng (PTW attack, 15,477 IVs), decrypted correctly 100%
Decrypted capture filename/hash	file_working-dec (SHA-256: 167c91994c269777f9048227deb89882caf3fc3763977f2059604f9a6a40b04)
Host IP and MAC	192.168.0.15 / f0:f6:1c:68:96:7c
Server IP and MAC	198.90.20.111 / reached via gateway 00:26:66:55:97:d4
Key protocols observed	ARP, DHCP, DNS, IGMPv2, MDNS, TCP, TLSv1/TLSv1.2, HTTP
Recovered images/HTML	JPG, GIF, PNG and HTML objects recovered via tshark HTTP object export and foremost carving; hashed in reports/exported_file_hashes.txt
Optional router passphrase	Protected appendix only - optional Part G not assigned for this delivery
Security conclusion	WEP40's 24-bit IV space and RC4 keystream reuse permitted full key recovery in well under two minutes; WEP should be retired in favour of WPA2-AES/CCMP or WPA3

10.3 Command Reference

```
mkdir -p ~/SBT-DF203-Lab9/{evidence,working,exported,reports,screenshots,scripts}
wget -4 -O evidence/file.xz '<historical CodeGate CTF 2015 Good Crypto capture URL>'
sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
cp --preserve=timestamps evidence/file.xz working/file_working.xz
unxz -k working/file_working.xz
capinfos working/file_working | tee reports/capinfos.txt
tshark -r working/file_working -q -z io,phs
tshark -r working/file_working -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv
aircrack-ng working/file_working
airdecap-ng -w A43DF6F374 working/file_working
tshark -r working/file_working-dec -q -z endpoints,ip
tshark -r working/file_working-dec --export-objects http,exported/http_objects
foremost -i working/file_working-dec -o exported/foremost
```

Academic Integrity Declaration: I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.